EXPERIMENT - 12

AIM:

To implement "echo client server" using TCP/UDP Sockets Program

CODE: (TCP)

```
import socket
import threading

# --- SERVER ---
def start_server():
    server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    server.bind(('127.0.0.1', 5000))
    server.listen(1)
    print("[SERVER] Listening on 127.0.0.1:5000")
    conn, addr = server.accept()
    print(f"[SERVER] Connected by {addr}")
    while True:
        data = conn.recv(1024).decode()
        if not data or data.lower() == 'quit':
            print("[SERVER] Connection closed by client.")
            break
        print(f"[SERVER] Received: {data}")
        conn.send(f"Server received: {data}".encode())
    conn.close()
    server.close()

# --- CLIENT ---
def start_client():
    client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] Connected to server")
```

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```
While True:
    msg = input("Enter message (or 'quit' to exit): ")
    client.send(msg.encode())
    if msg.lower() == 'quit':
        break
    response = client.recv(1024).decode()
    print(f"[SERVER RESPONSE] {response}")
client.close()

# MAIN
if __name__ == "__main__":
    # Run server in a background thread
    server_thread = threading.Thread(target=start_server, daemon=True)
    server_thread.start()
    # Run client in main thread
    start_client()
```

OUTPUT:

```
[SERVER] Listening on 127.0.0.1:5000
[CLIENT] connected to server
[SERVER] connected by ('127.0.0.1', 53458)
Enter message (or 'quit' to exit):
Hello
[SERVER] Received : HELLO
[SERVER RESPONSE] Server received : Hello
Enter message (or 'quit' to exit): quit
[SERVER] connection closed by client
```

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UDP :

```
import socket
import threading
# -- SERVER --
def UDP_Server():
    S = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    s.bind(('127.0.0.1', 12000))
    Print ("[SERVER] Running on 127.0.0.1:12000")
While True:
    data, addr = S.recvfrom(1024)
    msg = data.decode()
    if msg.lower() == 'quit':
        Print ("[SERVER] client {addr} ended the connection.")
    break
    Print (f "[SERVER] Received from {addr}: {msg}")
    S.sendto (data, addr)
    S.close()
    Print ("[SERVER] Stopped.")
# --- CLIENT ---
def UDP (Client)
    C: socket.socket (Socket.AF-INET, Socket.SOCK-DGRAM)
Server = ('127.0.0.1', 12000)
While True:
    msg = input ("Enter message ('or 'quit' to stop): ")
    c.sendto (msg.encode(), Server)
    if msg.lower == 'quit':
Print ("[CLIENT] Closing connection...")
    break
data, = c.recv from (1024)
Print ("Echo from server:", data.decode())
c.close()
```

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```
# --- MAIN ---
if __name__ == "__main__":
    threading.Thread(target=udp_server, daemon=True).start()
    udp_client()
```

OUTPUT :

```
Enter message (or 'quit' to stop): [SERVER] Running on 127.0.0.1:12000
Hi
[SERVER] Received from ('127.0.0.1', 49870): Hi
Echo from server : Hi
Enter message (or 'quit' to exit) : quit
[CLIENT] Stopped
[SERVER] Stopped
```

RESULT :

Thus the Program to implement echo client server using TCP/UDP socket was executed successfully